

Product variety multiplies the stock needed for the same service

Part 3 · Why is it difficult? — solution report with data, method, results and charts

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1. Part 3 · Why is it difficult? — Product variety multiplies the stock needed for the same service

1. Product variety multiplies the stock needed for the same service

$$SS_{total}(n) = \sqrt{n} \times SS_{total}(1)$$

1,234 units

SS with 1 variant

\$63,908/yr

Extra holding cost at max variety

9,870 units (×8.0)

SS with 64 variants

95.0% / 95.0%

Target / simulated CSL

The cost of variety: the square-root law

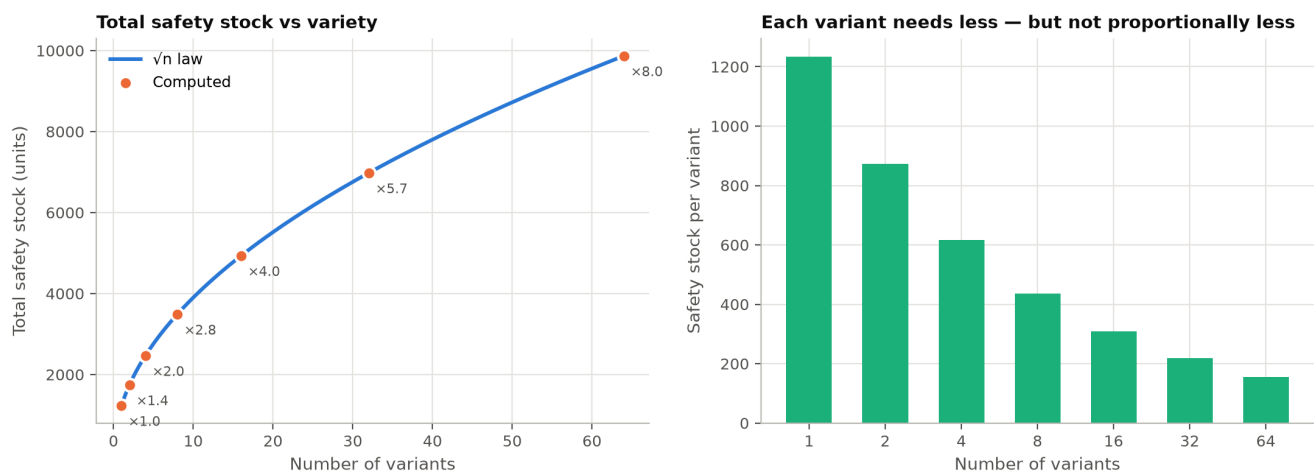


Figure 1.1 — Total demand 1,000/day, CV 25%, lead time 9 days, $z = 1.645$.

Safety stock as variety grows.

n variants	mean daily demand per variant	sigma per variant	SS per variant	SS total	SS total index	simulated CSL	annual holding cost SS
1	1,000.0	250.0	1,233.8	1,234	1.00	95.5%	9,130
2	500.0	176.8	872.4	1,745	1.41	94.6%	12,911
4	250.0	125.0	616.9	2,468	2.00	95.1%	18,260
8	125.0	88.4	436.2	3,490	2.83	95.0%	25,823
16	62.5	62.5	308.4	4,935	4.00	95.0%	36,519
32	31.2	44.2	218.1	6,979	5.66	95.0%	51,646
64	15.6	31.2	154.2	9,870	8.00	94.9%	73,038

- Promotions create short demand spikes → anticipation stock or capacity costs (Parts 2.4, 6.6).
- Variety should be priced with its inventory consequence; *postponement* (customising late) pools demand.